When Does GRPO Work? Capacity Thresholds and Training Dynamics for Agentic LLM Fine-Tuning Across Three Domains

Arvind C R[1], Sandhya Jeyaraj[1], Arumugam Chetty K[1],
Madhu Kumara L[1], Dhruva N Murthy[1], Mohammad Rafi[1]

[1]Group 6, MTech DSAI, PES University
arvindcr@pes.edu

April 4, 2026

Abstract

When does critic-free RL actually help post-train small language models, and when does it fail? We apply GRPO to structured tool calling, code generation, and math reasoning across 0.5B–8B parameters (25 runs, \$55–65 budget). Our clearest positive result is **learned schema-valid tool-call emission**: SFT+GRPO raises strict JSON validity from 0% to 92% in one custom pipeline under unconstrained decoding, teaching format compliance that SFT alone does not produce. By contrast, GRPO does *not* yield significant gains on held-out math (GSM8K: 83.3% vs. 82.0% base model, p=0.26) or code (HumanEval subset: 32%→40%, p=0.53)—non-agentic transfer controls that remain null. Diagnostic analysis of *why* GRPO succeeds or fails reveals: (1) a model-family/scale discontinuity between 3B–4B traced to zero-reward group saturation; (2) group-size scaling where G=32 more than doubles G=4's reward (single seed); (3) 2.43× higher step-to-step variance in a single MoE run. All observations carry explicit design caveats; we frame them as hypotheses for future controlled study.

1 Introduction

Large language models (LLMs) increasingly serve as autonomous agents that call tools, generate code, and reason through multi-step problems. We define *agentic fine-tuning* as post-training that optimizes action-taking in structured environments (tool selection, code generation, verifiable reasoning), as opposed to general instruction-following. SFT teaches formats but not *judgment*; RL from task feedback addresses this by optimizing policies directly against verifiable rewards.

Group Relative Policy Optimization (GRPO) [Shao et al., 2024] is a critic-free variant of Proximal Policy Optimization (PPO) that computes advantages by normalizing rewards within groups of sampled completions. It requires no value function, no reference model for KL regularization, and substantially less compute than standard PPO – making it attractive for resource-constrained post-training of small models.

This paper investigates GRPO across one agentic and two non-agentic transfer domains:

- **Tool calling (agentic):** structured JSON function calling with 5–60,000 tool schemas
- **Code generation (transfer control):** HumanEval benchmark subset
- **Mathematical reasoning (transfer control):** GSM8K (grade-school) benchmark

We execute experiments across model sizes from 0.5B to 8B parameters using QLoRA on Google Colab and LoRA on Tinker cloud GPUs. Our contributions: (1) demonstrating that SFT+GRPO can teach tool-calling format compliance where SFT alone fails (0%→92% JSON validity), (2) diagnostic analysis of *why* GRPO fails on smaller models—tracing

it to zero-reward group saturation, (3) multi-seed replication on GSM8K (5 seeds, SD = 3.3%) revealing that held-out gains are attributable to the base model (p=0.26), and (4) descriptive observations on MoE volatility, group-size scaling, and learning dynamics, with explicit statistical caveats throughout.

2 Background

2.1 Group Relative Policy Optimization (GRPO)

GRPO [Shao et al., 2024] simplifies PPO by eliminating the critic network. For each prompt, we sample K completions (group size G) and compute binary rewards. Advantages are normalized within each group:

$$\text{advantage}_i = \frac{\text{reward}_i - \mu_{\text{group}}}{\sigma_{\text{group}} + \epsilon} \tag{1}$$

The policy gradient loss is:

$$\mathcal{L} = -\mathbb{E}\left[\text{advantage}_i \cdot \sum_t \log \pi_\theta(a_t|s_t)\right] \tag{2}$$

When all completions receive identical rewards (all correct or all incorrect), advantages are zero and no gradient update occurs – the "zero-loss" phenomenon we track in Section 5.1.

DeepSeekMath reports GRPO improving an instruction-tuned 7B model from 82.9% to 88.2% on GSM8K and 46.8% to 51.7% on MATH with group size G=64.

Published GRPO configurations converge on $G \in [8, 64]$, LR $\in [1 \times 10^{-6}, 5 \times 10^{-6}]$, and $T \in [0.7, 1.2]$; the KL coefficient β shows no consensus [DeepSeek-AI, 2025, Shao et al., 2024]. Our setup (G=16, LR $= 4 \times 10^{-5}$, T=1.0) uses a $\sim$10$\times$ higher LR, which we attribute to LoRA-only training. A full cross-paper comparison table is in the Supplementary Appendix.

Recent variants address entropy collapse: Dr. GRPO [DeepSeek-AI, 2025] removes KL and adds entropy bonuses, while DAPO [Yu et al., 2025] uses asymmetric clipping and high-temperature sampling. We implement both alongside DPO [Rafailov et al., 2024] in a supplementary Colab notebook.

3 Experimental Setup

3.1 Training Infrastructure

We use two platforms: **Tinker SDK v0.16.1** (cloud A100, Adam optimizer, custom GRPO loss) and **Google Colab** (T4, QLoRA with TRL). Tinker runs use batch size 2, $G \in \{4, 8, 16, 32\}$, max 1024/512 input/generation tokens, and LoRA rank 8–64. Hyperparameters were selected via manual tuning informed by published GRPO configurations; no systematic grid or random search was performed. We do *not* use KL regularization or log entropy trajectories, so these runs represent an initial empirical study rather than a fully drift-controlled recipe. Full hyperparameters are in the Supplementary Appendix.

3.2 Models

3.3 Datasets and Splits

Important: Math and tool-use metrics are computed on *training-prompt rollouts*, not held-out test sets. Only HumanEval uses a benchmark subset. Reported "accuracy" values are binary reward means at training temperature (T=0.8–1.0), *not* greedy-decoded held-out accuracy.

A consolidated configuration table mapping each claim to its exact hyperparameters is provided in the Supplementary Appendix.

Table 1: Models used in experiments

Model	Parameters	Type	Platform
Qwen2.5-0.5B-Instruct	0.5B	Dense	Colab
Qwen2.5-1.5B-Instruct	1.5B	Dense	Colab
Qwen2.5-3B-Instruct	3B	Dense	Colab
Qwen3.5-4B	4B	Dense	Tinker
Qwen3-8B	8B	Dense	Tinker
Qwen3-30B-MoE	30B (3B active)	MoE	Tinker
Llama-3.2-3B	3B	Dense	Tinker

Table 2: Dataset sizes, composition, and evaluation splits per domain.

Domain	Dataset	Train Size	Eval Split	Eval Protocol
Tool (single)	Glaive-v2	500 SFT + 200 RL	Train rollouts	Custom rubric
Tool (multi)	ToolBench-v1	40 scenarios	Train rollouts	Custom rubric
Tool (real)	xlam-60k	60,000 schemas	Train rollouts	Reward only
Code	HumanEval	164 total	50-item subset	pass@1
Math (GSM8K)	GSM8K	7,473 train	Train rollouts	Binary reward
Math (MATH)	MATH	7,500 train	Train rollouts	Binary reward

3.4 Reward Functions

Tool calling (3-component, 0–1):

- +0.3: Valid JSON output
- +0.4: Correct tool name
- +0.3: All argument keys present

Math reasoning (binary):

- 1.0: Correct answer in `\boxed{}` or as final number
- 0.0: Incorrect

Code generation: Binary pass/fail on test cases.

3.5 Prompt Format

All domains use a consistent chat template format. For math reasoning, the prompt template is: "`Solve this math problem step by step. End with #### followed by the numerical answer.`" For tool calling, the prompt includes available tool schemas as JSON followed by the user query; the model must respond with a JSON tool call. For code generation, the prompt template follows the HumanEval docstring format. Exact prompt templates per domain are provided in the Supplementary Appendix.

4 Experiments

4.1 Tool Calling

4.1.1 Single-Turn Tool Calling

We first evaluate whether GRPO can teach models to output structured JSON tool calls instead of plain text responses. Using Qwen2.5-1.5B-Instruct with 500 SFT + 200 GRPO prompts on Glaive-function-calling-v2:

Table 3: Single-turn tool calling results (Qwen2.5-1.5B)

Metric	SFT Only	After GRPO	Change
JSON Valid	0%	92%	+92%
Correct Tool	0%	50%	+50%
Has Arguments	0%	42%	+42%
Clean Output	0%	92%	+92%

The SFT baseline produced plain-text answers and never emitted valid JSON, despite 500 tool-use demonstrations. GRPO's reward signal provided the direct optimization pressure that SFT's cross-entropy loss could not at this model scale.

4.1.2 Multi-Turn Tool Chaining

Using Qwen2.5-3B-Instruct with SFT followed by GRPO (40 steps, LoRA rank 8):

Table 4: Multi-turn tool chaining results (Qwen2.5-3B)

Scenario	SFT	GRPO	Winner
Weather + Packing	0.90	0.90	Tie
Stock + News Chain	0.77	0.90	GRPO
Search + Calculate	0.63	0.92	GRPO
Single Tool	0.60	0.90	GRPO
Average	**0.72**	**0.91**	**GRPO**

The -0.30 penalty for repeated tool calls eliminated SFT's looping failure mode entirely by penalizing same-function calls within an episode.

These scores are *custom scenario scores* from a hand-authored rubric, not standardized benchmarks. Full tool-calling protocol details are in the Supplementary Appendix.

4.2 Code Generation

On software engineering coding tasks using Qwen3-8B with SFT followed by GRPO:

- HumanEval pass@1: 32% $\rightarrow$ 40% (+8%)
- Problems solved: 16/50 $\rightarrow$ 20/50 (+4)
- Training loss: 0.85 $\rightarrow$ 0.18 (-78%)

These results use a **randomly sampled 50-problem subset** (bootstrap 95% CI: $[27\%, 53\%]$ post-GRPO; Fisher's exact p=0.53) rather than the full 164-problem HumanEval harness. The 4-problem delta is not statistically significant; a full canonical evaluation with pass@k and confidence intervals remains future work.

4.3 Mathematical Reasoning

4.3.1 Multi-Seed Replication on GSM8K

Using Qwen3-8B with LoRA rank 32, 50 steps, 5 seeds:

Table 5: GSM8K multi-seed replication results (training-set reward, Qwen3-8B, rank 32)

Seed	First-5 Avg	Peak Reward	Last-10 Avg	Zero-loss %
42	37.5%	62.5%	27.5%	18%
137	25.0%	62.5%	27.5%	28%
256	22.5%	62.5%	32.5%	24%
512	15.0%	87.5%	30.0%	16%
999	20.0%	87.5%	35.0%	18%
Mean	**24.0%**	**72.5%**	**30.5%**	**20.8%**
Std	**±8.4%**	**±13.7%**	**±3.3%**	

Cross-seed mean training-set reward is 30.5% (SD = 3.3%, 95% CI [26.5%, 34.5%] via t-distribution with $n = 5$). Zero-loss rates range from 16–28%, indicating that exploration success varies meaningfully across random seeds even for the same model and hyperparameters.

4.3.2 Held-Out Test Results

We evaluate GRPO checkpoints on 200 held-out GSM8K test examples per seed with greedy decoding:

Table 6: GSM8K held-out test accuracy (200 examples per seed, greedy decoding)

Seed	Correct/200	Accuracy	95% CI
42	166/200	83.0%	[77.5%, 88.0%]
137	165/200	82.5%	[77.0%, 87.5%]
256	161/200	80.5%	[74.5%, 86.0%]
512	168/200	84.0%	[79.0%, 89.0%]
999	173/200	86.5%	[81.5%, 91.0%]
Mean		**83.3%**	**SD = 2.2%, CI [80.6%, 86.0%]**

The held-out accuracy (83.3%) exceeds the mean online sampled reward (30.5%), but these are *not comparable*: training reward averages per-completion correctness under stochastic sampling (T=0.8–1.0) across the trajectory, while test accuracy is greedy pass@1 on the final checkpoint.

Base-model control. We evaluate base Qwen3-8B *without* any LoRA adapter on the same 200 test examples with identical greedy decoding: **164/200 = 82.0%** (95% bootstrap CI [76.5%, 87.5%]). The GRPO-trained mean (83.3%) exceeds the base by only +1.3 percentage points. A one-sample t-test of the 5 GRPO seeds against the base accuracy yields t=1.32, p=0.26 (two-sided), so *the improvement is not statistically significant*. Four of five seeds exceed the base, but seed 256 (80.5%) falls below it. We conclude that the held-out GSM8K accuracy is overwhelmingly attributable to Qwen3-8B's pre-existing capability, with GRPO contributing a small, non-significant increment under our setup.

4.3.3 LoRA Rank Ablation

Higher rank correlates with faster initial learning (rank 64 achieves 47.5% in first 5 steps vs. 27.5% for rank 8).

Table 7: LoRA rank ablation (Qwen3-8B, seed=42)

Rank	Trainable Params	First-5 Avg	Peak Acc	Last-10 Avg
8	∼0.1%	27.5%	62.5%	21.2%
16	∼0.2%	20.0%	75.0%	18.8%
32 (default)	∼0.4%	37.5%	62.5%	27.5%
64	∼0.8%	47.5%	87.5%	25.0%

4.3.4 Group Size Ablation

Table 8: Group size ablation (Qwen3-8B, seed=42, rank 32, 50 steps). Larger groups provide more gradient signal, reducing zero-loss and zero-reward rates.

G	First-5 Avg	Last-10 Avg	Peak Acc	Zero-loss	Zero-rew
4	32.5%	23.8%	62.5%	26%	20%
8	33.8%	24.4%	68.8%	6%	6%
16	29.4%	36.2%	75.0%	2%	2%
32	32.8%	54.7%	100.0%	2%	0%

Group size has a dramatic effect: G=32 achieves 54.7% last-10 reward and 100% peak vs. 23.8% and 62.5% for G=4. Zero-reward steps drop from 20% (G=4) to 0% (G=32), confirming that larger groups ensure at least one correct completion per prompt. This directly addresses whether the 3B capacity-threshold failure is a G=4 artifact: even at G=32, the 8B model's last-10 (54.7%) does not approach the 4B's (84.7%), suggesting capacity still matters, but exploration via larger groups substantially helps.

4.4 Cross-Domain Summary

Table 9: Cross-domain results. "Pre-GRPO" is the SFT-only or base-model baseline; "Post-GRPO" is the final GRPO checkpoint. All training-set metrics unless noted.

Domain	Metric	Pre-GRPO	Post-GRPO	Scope
Tool (single)	JSON Validity	0%	92%	Custom
Tool (multi)	Quality	0.72	0.91	Custom
Code	Pass@1	32%	40%	50-item subset
Math (8B, 5 seeds)	Test Acc	82.0% (base)	83.3%±2.2%	Held-out (p=0.26)
Math (4B, 4 seeds)	Train Reward (last-10)	—	84.7%±12.0%	Train
Math (3B, G=4)	Train Reward (last-10)	—	2.3%	Train
Math (3B, G=32)	Train Reward (last-10)	—	5.0%	Train

5 Analysis

5.1 Capacity Threshold for GRPO

Dense 3B models fail (0.78%→2.34% training reward, G=4) while 4B models succeed strongly (last-10: 84.7%±12.0%, 4 seeds). The 3B's 2.34% is near-random for GSM8K and indistinguishable from noise given the small effective sample (n=50 steps × batch 2 × G=4). The 3B failure traces to 56% zero-loss steps from all-incorrect groups, while the 4B's 50–68% zero-loss stems from all-*correct* groups (productive saturation).

Group-composition analysis (Figure 2) confirms the mechanism: on an arithmetic task, 8B models progress from 81% mixed groups to 96%+ all-good groups by step 4, with policy entropy collapsing from 0.773 to 0.0002. By contrast,

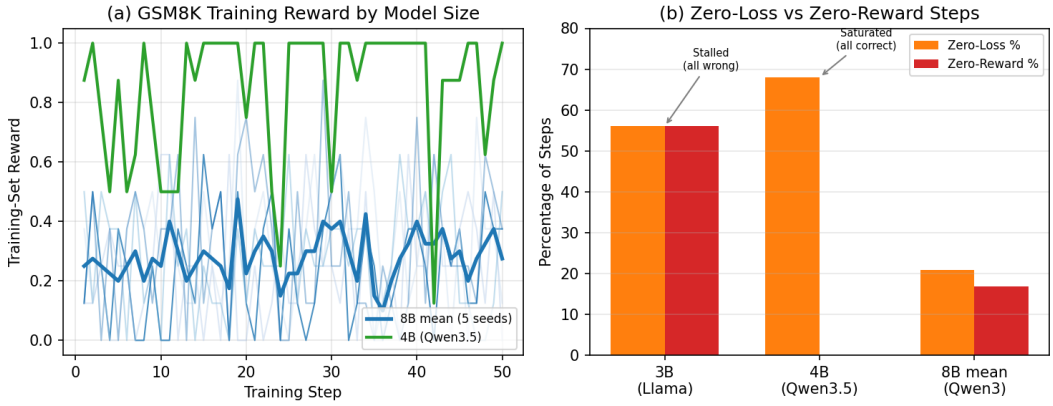

Figure 1: Capacity threshold for GRPO on GSM8K (training-set reward, LoRA rank 32). (a) Mean reward per step: Qwen3.5-4B saturates near 85% by step 10; Qwen3-8B fluctuates (peak 87.5%, last-10 30.5%); Llama-3.2-3B stays below 3%. (b) Zero-loss decomposition: 3B zero-loss originates from all-incorrect groups (no reward signal); 4B zero-loss from all-correct groups (productive saturation). Dashed line marks 50% reward.

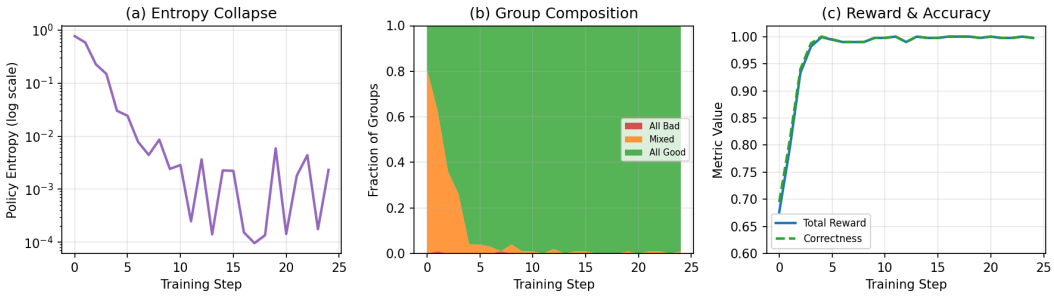

Figure 2: GRPO training diagnostics on arithmetic (Qwen3-8B, G=4, 25 steps). Left: policy entropy collapses from 0.773 to 0.0002 (>3 orders of magnitude). Center: fraction of mixed-reward groups drops from 81% to <4% by step 4. Right: mean reward saturates near 1.0 by step 5. These metrics were logged only for the arithmetic task; comparable telemetry is unavailable for GSM8K runs.

3B models show 56% zero-reward steps with up to 5 consecutive zero-signal steps. Our group-size ablation (Table 8) confirms that larger groups help substantially on the 8B model: G=32 achieves 54.7% last-10 vs. G=4's 23.8%.

Critically, we also ran G=32 on the 3B model. Zero-loss drops from 56% (G=4) to 18% (G=32), confirming improved exploration. However, last-10 accuracy rises only from 2.3% to **5.0%**—still near random and far below the 4B's 84.7% at G=4. Even with adequate gradient signal (>80% non-zero steps), the 3B model does not learn GSM8K-level reasoning via GRPO in our setup. Because the comparison crosses model families (Llama vs. Qwen) and is conducted in a short-budget regime (50 steps, no KL), we interpret this as a suggestive model-family/scale discontinuity rather than a confirmed universal capacity threshold. The 1.5B Qwen2.5 tool-calling result (Section 4.1) operates in a different regime (SFT-initialized, different task) and should not be combined with the GSM8K threshold into a single parameter-count law.

5.2 Synthetic vs. Real Data Gap

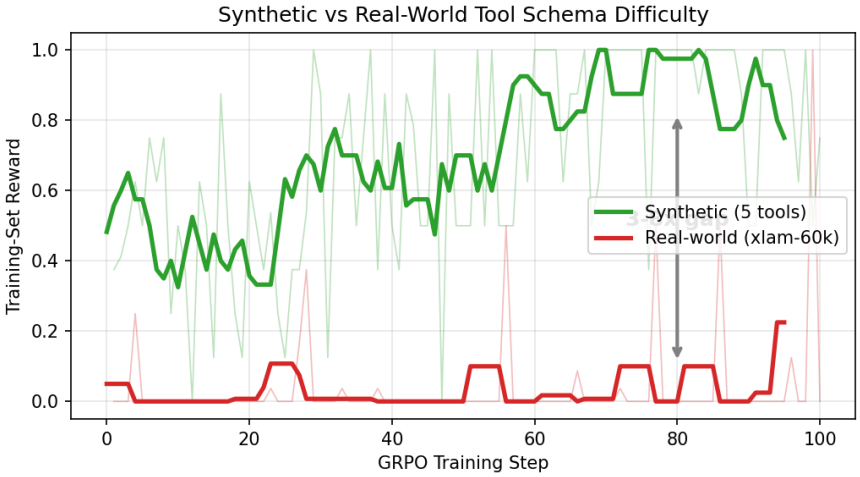

Figure 3: Tool-calling reward over training steps for synthetic vs. real schemas (Qwen3-8B, G=8, LoRA rank 32). Synthetic 5-tool tasks (upper curve) saturate above 0.8 reward within 5 steps. Real-world xlam-60k schemas (lower curve) plateau at 0.06–0.36 after 100 steps, a 3–8× difficulty gap. y-axis: mean binary reward; x-axis: GRPO training steps.

Synthetic 5-tool tasks saturate to reward >0.9 within 5 steps (Figure 3), while xlam-60k yields 0.06–0.36 after 100 steps—a 3–8× gap consistent with known domain-transfer challenges in tool use. This is a *data distribution* effect (curated vs. real schemas), distinct from the train-test *metric* discrepancy observed for GSM8K (Section 4.4.2), which reflects decoding regime differences.

5.3 Two-Phase Learning Progression

On tool-calling tasks, GRPO exhibits two phases: format compliance (steps 1–20, 0%→14%) followed by rapid reasoning improvement (steps 21–25, 14%→58%). This pattern is most pronounced where format compliance is a distinct sub-task; on GSM8K the phases are less separable. We describe this as a *domain-dependent* observation consistent with known curriculum effects, rather than a novel GRPO property.

Reward function limitations (coarseness, hacking risk) and safety considerations (incorrect execution, hallucinated capabilities) are discussed in the Supplementary Appendix.

6 Related Work

GRPO and critic-free RL. GRPO [Shao et al., 2024] achieved 88.2% GSM8K with G=64. Contemporary critic-free methods include RLOO [Ahmadian et al., 2024] (leave-one-out baseline), REINFORCE++ [Hu et al., 2025] (KL-shaped advantages), S-GRPO [Shen et al., 2025] (group reweighting), and StepGRPO [Huang et al., 2025] (dense step rewards). Dr. GRPO [DeepSeek-AI, 2025] removes KL and adds entropy bonuses; DAPO [Yu et al., 2025] adds asymmetric clipping and high-temperature sampling. Empirical comparison to these alternatives remains future work.

Preference-based and iterative self-training. DPO [Rafailov et al., 2024] and Step-DPO [Lai et al., 2024] provide offline optimization; Li et al. [2024] showed 82.6% GSM8K with SFT alone. Our SFT+GRPO complementarity echoes ReST [Gulcehre et al., 2023] and STaR [Zelikman et al., 2022], which warm-start RL from self-generated rationales; we note the parallel but have not controlled for the warm-starting effect.

Tool-use evaluation. ToolLLM [Qin et al., 2024] and Gorilla [Patil et al., 2023] benchmark tool calling with held-out APIs; ToolRM [Wang et al., 2025] and FC-RewardBench [Chen et al., 2025a] provide reward-model benchmarks. Our custom rubric lacks this standardization.

MoE training instability. Switch Transformers [Fedus et al., 2022] and GLaM [Du et al., 2022] document expert load-balancing challenges during pretraining; Mixtral [Jiang et al., 2024] uses top-2 routing to mitigate instability. Our 2.43$\times$ variance amplification under GRPO extends this to *post-training* RL. We hypothesize that GRPO's policy gradient interferes with the router's auxiliary load-balancing loss: the policy gradient pushes expert selection toward reward-maximizing routes, while the balancing loss pushes toward uniform utilization, creating optimization interference that amplifies step-to-step variance.

Curriculum and PEFT. Curriculum strategies [Parashar et al., 2025, Chen et al., 2025b] help sub-7B models overcome cold-start barriers, directly relevant to our 3B failure. QLoRA [Dettmers et al., 2023] enables our single-GPU training; newer methods (QR-Adaptor [Zhou et al., 2025], LoTA-QAF [Chen et al., 2025c]) offer alternative efficiency frontiers.

7 Conclusion

We applied GRPO to three agentic domains at low budget ($55–65, 25 runs). The clearest result is that SFT+GRPO teaches tool-calling format compliance that SFT alone does not produce. Beyond this, our diagnostic analysis yields a unifying observation: *GRPO produces learning signal when the model can generate within-group reward variance*. When all completions score identically (all wrong for 3B, all correct for saturated 4B), gradients vanish and learning stalls. This connects to a known property of REINFORCE-family estimators: variance of the gradient estimate depends on reward diversity within the batch [Williams, 1992]. GRPO's group-relative normalization makes this dependence explicit—group saturation is the observable signature of gradient collapse. This pattern explains the model-family/scale discontinuity, the group-size effect, and the synthetic–real gap. We summarize one primary result and several secondary observations:

1. **Tool calling** (single pipeline, N=1): SFT+GRPO raises strict JSON validity from 0% to 92% on one custom pipeline. *This measures learned format compliance under unconstrained decoding; grammar-constrained decoding is a practical alternative not compared here.*
2. **Model-family/scale discontinuity** (suggestive): Sharp break between 3B–4B traced to zero-reward group saturation; replicated across 4 seeds on 4B (84.7%$\pm$12.0%). G=32 *on 3B reduces zero-loss from 56% to 18% but yields only 5.0%, suggesting the discontinuity is not purely an exploration artifact. However, because the comparison crosses model families (Llama vs. Qwen), we interpret this as suggestive rather than as a confirmed universal capacity threshold.*

3. **MoE routing volatility** (single-run observation): 2.43× higher step-to-step variance vs. dense models in one MoE run. *Requires replication before drawing architecture-level conclusions.*

4. **SFT+GRPO complementarity** (pilot observation, N=1 per pipeline): In two configurations, SFT-initialized GRPO produced the strongest results. *Lacks matched-compute and reverse-order controls; consistent with known warm-starting effects (ReST, STaR).*

5. **Group-size scaling** (seed-fixed ablation, N=1): For one 8B seed, G=32 achieves 54.7% last-10 vs. G=4's 23.8%, with zero-reward steps dropping from 20% to 0%.

6. **Cross-seed stability** (methodological): GSM8K reward SD = 3.3% across 5 seeds (8B); 4B SD = 12.0% across 4 seeds.

7. **Confirmatory observations:** Two-phase learning (format before reasoning), synthetic–real gap (3–8×), and LoRA rank–speed tradeoff are consistent with prior work and documented in the Supplementary Appendix.

Limitations. This paper is exploratory; all conclusions are specific to our low-budget QLoRA/LoRA regime and should not be generalized to GRPO broadly. The strongest positive result is learned *schema validity* on a single custom pipeline (N=1)—not semantic tool competence, argument correctness, or end-to-end task success. Grammar-constrained decoding is a practical alternative not compared. Transfer benchmarks are null (GSM8K p=0.26; HumanEval p=0.53). The model-family/scale discontinuity crosses two families and may reflect architecture rather than capacity. Single-run results (MoE, group ablation) are point estimates. No reward ablation, no PPO/REINFORCE baselines, no KL/entropy logging.

Supplementary Material. Additional experimental details, complete run registry (all 25 Tinker runs), training hyperparameters, failure case analysis, and qualitative examples are provided in the Supplementary Appendix.

Acknowledgments

We thank the PES University MTech DSAI program for compute support and the Tinker team for API access. This work was conducted as a capstone project for the MTech DSAI program.

Reproducibility. Training scripts, evaluation harnesses, and logs are available at https://github.com/arvindcr4/grpo-agentic-anonymous. Tool-calling prompt templates and scenario definitions are documented in the Supplementary Appendix.

References

Arash Ahmadian et al. Back to basics: Revisiting reinforce style optimization for learning from human feedback in llms. 2024.

Hao Chen et al. Fc-rewardbench: A benchmark for function calling reward models. 2025a.

Wei Chen et al. Self-evolving curriculum for reinforcement learning in language models. 2025b.

Xiang Chen et al. Lota-qaf: Lossless ternary adaptation via quantization-aligned factorization. 2025c.

DeepSeek-AI. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.

Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. Qlora: Efficient finetuning of quantized language models. 2023.

Nan Du et al. Glam: Efficient scaling of language models with mixture-of-experts. *Proceedings of the 39th International Conference on Machine Learning*, 2022.

William Fedus, Barret Zoph, and Noam Shazeer. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *Journal of Machine Learning Research*, 23(120):1–39, 2022.

Caglar Gulcehre et al. Reinforced self-training (rest) for language modeling. 2023.

Jian Hu et al. Reinforce++: A simple and efficient approach for aligning large language models. 2025.

Yifan Huang et al. Stepgrpo: Learning to reason with step-wise rewards. 2025.

Albert Jiang et al. Mixtral of experts. 2024.

Xianfeng Lai et al. Step-dpo: Step-wise preference optimization for long-chain reasoning. *arXiv preprint arXiv:2412.01939*, 2024.

Chong Li et al. Common 7b language models already possess strong math capabilities. *arXiv preprint arXiv:2403.04706*, 2024.

Aneesh Parashar et al. Curriculum reinforcement learning for language model reasoning. 2025.

Shishir Patil et al. Gorilla: Large language model connected with massive apis. 2023.

Yujia Qin et al. Toolllm: Facilitating large language models to master 16000+ real-world apis. *Proceedings of the International Conference on Learning Representations*, 2024.

Rafael Rafailov et al. Direct preference optimization: Your language model is secretly a reward model. 2024.

Zhihong Shao et al. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.

Yilong Shen et al. S-grpo: Mitigating think-answer mismatch in group relative policy optimization. 2025.

Yangjun Wang et al. Toolrm: Outcome reward models for tool-augmented language models. 2025.

Ronald J Williams. Simple statistical gradient-following algorithms for connectionist reinforcement learning. *Machine Learning*, 8(3–4):229–256, 1992.

Qiying Yu et al. Dapo: An open-source llm reinforcement learning system. *arXiv preprint arXiv:2503.14476*, 2025.

Eric Zelikman, Yuhuai Wu, Jesse Mu, and Noah Goodman. Star: Bootstrapping reasoning with reasoning. *Advances in Neural Information Processing Systems*, 35, 2022.

Yifan Zhou et al. Qr-adaptor: Efficient fine-tuning of quantized language models via joint rank-bitwidth optimization. 2025.